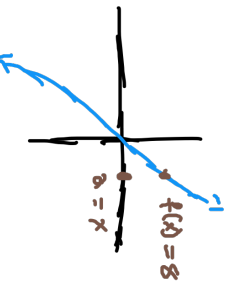


Week 13: Inverse functions

Idea: f^{-1} 'undoes' f

$$f(x) = x^3$$

$$f^{-1}(x) = x^{1/3}$$



$$f^{-1}(8) = 2$$

$$f(2) = 8$$

$$f^{-1}(8) = 2$$

"the x s.t. $f(x) = 26$ " $= f^{-1}(26) = 26^{1/3} = 2.995...$

FE solve the eqn $f(x) = 26$ for x

$$(*) \quad (f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(x^3) = (x^3)^{1/3} = x = \text{id}_{\mathbb{R}}(x)$$

$$f^{-1} \circ f = \text{id}_{\mathbb{R}}$$

$$f(x) = x^3 = y \quad \text{+ solve for } x$$

$$x = y^{1/3}$$

"works", but isn't the defn

$\text{id}_A: A \rightarrow A$ "the" identity fcn

$$x \mapsto x$$

$$\text{id}_A(x) = x, \forall x \in A$$

Prop: Let $f: A \rightarrow B$ be any fcn. Then

$$(i) f \circ \text{id}_A = f$$

$$A \xrightarrow{\text{id}_A} A \xrightarrow{f} B$$

$$(ii) \text{id}_B \circ f = f$$

$$A \xrightarrow{f} B \xrightarrow{\text{id}_B} B$$

Proof. (i) $f \circ \text{id}_A$ and f have the same domain & codomain.

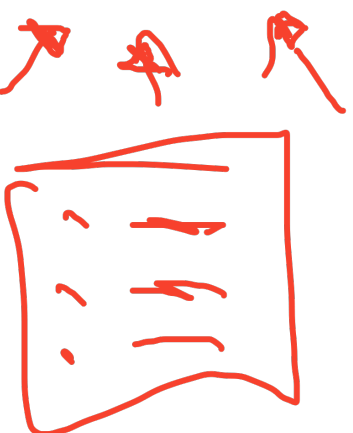
Let $a \in A$. Then $(f \circ \text{id}_A)(a) = f(\text{id}_A(a)) = f(a)$.
(ii) is similar.

Defn: We say that a fcn $f: A \rightarrow B$ is invertible if

$$\exists g: B \rightarrow A \text{ s.t. } f \circ g = \text{id}_B \\ g \circ f = \text{id}_A$$

When such a g exists, we call g an inverse of f and sometimes write $g = f^{-1}$.

Warnings: (i) $f^{-1} \neq \frac{1}{f}$



(ii) Not every f has an inverse!

$$A = B = \mathbb{R}$$

$$f: A \rightarrow A$$

$$x \mapsto x^3$$

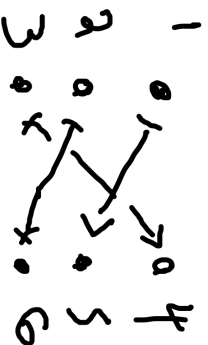
$$g: x \mapsto x^{1/3}$$

$$(g \circ f)(x) = g(f(x)) = g(x^3) = (x^3)^{1/3} = x = \text{id}_{\mathbb{R}}(x)$$

$$(f \circ g)(x) = f(g(x)) = f(x^{1/3}) = (x^{1/3})^3 = x = \text{id}_{\mathbb{R}}(x)$$

Example:

$$A \xrightarrow{f} B$$



what is the inverse of f ?

$$f \circ g = \text{id}_B$$

$$g \circ f = \text{id}_A$$

$$B \xrightarrow{g} A$$



$$(g \circ f)(1) = g(f(1)) = g(4) = 1$$

$$g \circ f)(z) = g(f(z)) = z$$

$$g(y) = z$$

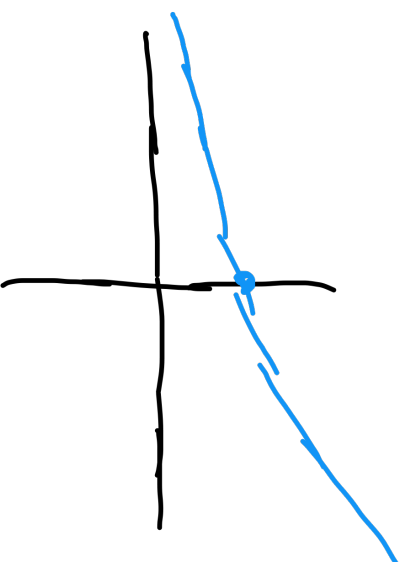
" $g(x)$ is the solution to $f(z) = x$ "

$$g(x) = z \iff x = f(z)$$

Example:

$$\mathbb{R} \xrightarrow{e} \mathbb{R}_{>0}$$

$$x \mapsto e^x$$



This has an inverse $\ln$

$$\mathbb{R}_{>0} \xrightarrow{\ln} \mathbb{R}$$

$$x \mapsto \ln x$$

$$\ln \circ e = \text{id}_{\mathbb{R}}$$

$$e \circ \ln = \text{id}_{\mathbb{R}_{>0}}$$



$$e^0 = 1$$

$\ln 1 = 0$ "the solution to $e^x = 1$ "

$$e^\pi = e^\pi \dots$$

$$\ln(e^\pi) = \pi$$

$$\ln(e^x) = x$$

$$e^{\ln x} = x$$

$$e^x = y \quad \Leftrightarrow$$

$$\ln y = x$$

$$\mathbb{R} \xrightarrow{f} \mathbb{R}$$

$$x \mapsto x^5 + 4x$$

$$f'(x) = 5x^4 + 4 \geq 4$$

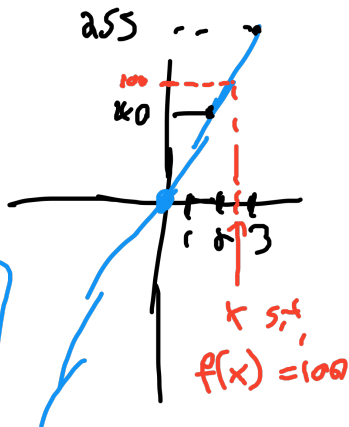
$\Rightarrow f$ is inj

$\exists$ v.t. $\Rightarrow$ surj

HS way

Solve $x^5 + 4x = y$ for x

... can't do this nicely



FACT f has an inverse g .

$$f(0) = 0 \quad g(0) = 0$$

$$f(1) = 5 \quad g(5) = 1$$

$$f(x) = y \Leftrightarrow g(y) = x$$

$$f(2) = 40 \quad g(40) = 2$$

$$f(x) = -5 \Leftrightarrow g(-5) = x$$

$$\Leftrightarrow$$

$$x^5 + 4x = -5 \Leftrightarrow x = -1$$

$$\text{Thus } g(-5) = -1$$

$$g(100) = x \Leftrightarrow f(x) = 100$$

$$\Leftrightarrow x^5 + 4x = 100$$

$$\Leftrightarrow x^5 + 4x - 100 = 0$$

By IVT, $\nearrow$ has a sol, b/c

$$f(2) = 40$$

$$f(3) = 3^5 + 4 \cdot 3 = 255$$

So $\exists x \in [2, 3]$ s.t. $f(x) = 100$

$$f^{-1}(100) = \text{this } x = 2 \dots$$

Sometimes f has no inverse

$$\mathbb{R} \xrightarrow{f} \mathbb{R} \\ x \mapsto x^2$$

① What is $f^{-1}(-1)$?

There is no input s.t. $f(x) = -1$

} Need set to have an inverse

Math hack: replace codomain of image.

② What is $f^{-1}(4)$?

$$f(2) = 4 \\ f(-2) = 4$$



Problem: is $f^{-1}(4) = 2$ or -2 ?

?

need inj to have an inverse

What about x "h"?

not def for $x < 0$
ambiguous (+ & - text)

THM: $f: A \rightarrow B$ has an inverse $\Leftrightarrow f$ is bijective.

Proof: " $\Rightarrow$ " Assume f has an inverse g .

(INS, WTS $\forall a, b \in A, f(a) = f(b) \Rightarrow a = b$)

Let $a, b \in A$. Suppose $f(a) = f(b)$. Then, $g(f(a)) = g(f(b))$.

Since $g = f^{-1}$, $g \circ f = \text{id}$, so $a = b$. ($(g \circ f)(a) = g(f(a)) = a$)

(S, WTS $\forall b \in B, \exists a \in A$ s.t. $f(a) = b$)

Let $b \in B$. Let $a = g(b)$. Then $f(a) = f(g(b)) = (f \circ g)(b) = \text{id}_B(b) = b$.

" $\Leftarrow$ " Assume f is bijective. Let's define $g: B \rightarrow A$ as follows.

Let $b \in B$. Since f is surj, $\exists a \in A$ s.t. $f(a) = b$. Since f is inj,

there is only one such a . Define $g(b) = a$. Then

$$(g \circ f)(a) = g(f(a)) = a. \quad \square$$

$$(f \circ g)(b) = f(g(b)) = b.$$

(1) USE THM $\left\{ \begin{array}{l} \text{to help w/ proofs} \\ \text{to help w/ counterexamples} \end{array} \right.$

(2) Given f , to find f^{-1} , "solve" the eqns

$$\begin{array}{l} f(y) = \text{val} \\ \text{got} = \text{val} \end{array} \quad \text{for } y$$

(2') If you have a guess for y ,
verify your guess by plugging y into

(i) x^2 not inv $\Rightarrow$ not invertible

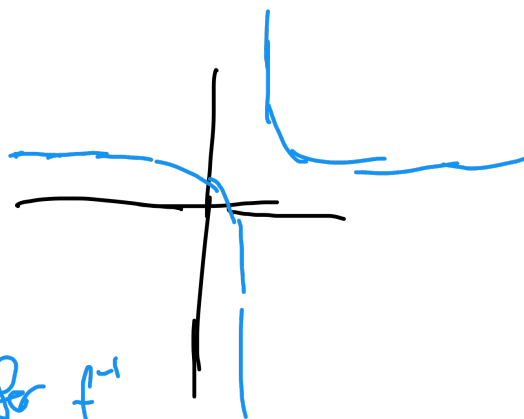
(ii) $p(x) = x^5 + x^3 + x$ as inv. by THM

b/c $b_{i,j}$,

(Still need to explain why $\&$ is $b_{i,j}$)

Ex: $\mathbb{R} - \{1\} \xrightarrow{f} \mathbb{R} - \{1\}$

$$x \mapsto \frac{x+1}{x-1}$$



How to find a formula for f^{-1}

$$f(x) = y \iff x = g(y)$$

" $g(y)$ is the x s.t. $f(x) = y$ "

$$\frac{x+1}{x-1} = y \quad \frac{2}{x-1} = y-1 \implies \frac{x-1}{2} = \frac{1}{y-1}$$

$$\parallel$$

$$\frac{x-1+2}{x-1} = 1 + \frac{2}{x-1} \Rightarrow$$

$$\Rightarrow x-1 = \frac{2}{y-1}$$

$$\Rightarrow x = \boxed{1 + \frac{2}{y-1}} = g(y)$$

TE $(f \circ g)(x) \stackrel{?}{=} x$

$$f\left(1 + \frac{2}{x-1}\right) = \frac{\left(1 + \frac{2}{x-1}\right) + 1}{\left(1 + \frac{2}{x-1}\right) - 1} \stackrel{\text{MAW}}{=} x$$

Example:

$$\mathbb{R}^2 \xrightarrow{f} \mathbb{R}^2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x-y \\ x+y \end{pmatrix}$$

$$\mathbb{R}^2 \xrightarrow{g} \mathbb{R}^2$$

$$\begin{pmatrix} s \\ t \end{pmatrix} \mapsto \begin{pmatrix} \frac{s+t}{2} \\ \frac{-s+t}{2} \end{pmatrix}$$

$$(g \circ f)\begin{pmatrix} x \\ y \end{pmatrix} = g\left(f\begin{pmatrix} x \\ y \end{pmatrix}\right)$$

$$= g\begin{pmatrix} x-y \\ x+y \end{pmatrix}$$

$$= \begin{pmatrix} \frac{(x-y) + (x+y)}{2} \\ \frac{-(x-y) + (x+y)}{2} \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$f \circ g$ vs id